When making a trigonometric substitution, we shall assume that θ is in the range of the corresponding inverse trigonometric function. Thus, for the substitution $x = a\sin\theta$, we have $-\pi/2 \le \theta \le \pi/2$. In this case, $\cos\theta \ge 0$ and

$$\begin{aligned}\sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2\sin^2\theta}\\ &= \sqrt{a^2(1-\sin^2\theta)}\\ &= \sqrt{a^2\cos^2\theta}\\ &= a\cos\theta.\end{aligned}$$

If $\sqrt{a^2 - x^2}$ occurs in a denominator, we add the restriction $|x| \neq a$, or, equivalently, $-\pi/2 < \theta < \pi/2$.

EXAMPLE ■ 1 Evaluate $\displaystyle\int \frac{1}{x^2\sqrt{16-x^2}}\,dx.$

SOLUTION The integrand contains $\sqrt{16-x^2}$, which is of the form $\sqrt{a^2-x^2}$ with $a = 4$. Hence, by (7.4), we let

$$x = 4\sin\theta \quad \text{for} \quad -\pi/2 < \theta < \pi/2.$$

It follows that

$$\begin{aligned}\sqrt{16-x^2} &= \sqrt{16 - 16\sin^2\theta}\\ &= 4\sqrt{1-\sin^2\theta} = 4\sqrt{\cos^2\theta} = 4\cos\theta.\end{aligned}$$

Since $x = 4\sin\theta$, we have $dx = 4\cos\theta\,d\theta$. Substituting in the given integral yields

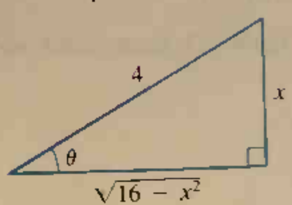
$$\begin{aligned}\int \frac{1}{x^2\sqrt{16-x^2}}\,dx &= \int \frac{1}{(16\sin^2\theta)\,4\cos\theta}\,4\cos\theta\,d\theta\\ &= \frac{1}{16}\int \frac{1}{\sin^2\theta}\,d\theta\\ &= \tfrac{1}{16}\int \csc^2\theta\,d\theta\\ &= -\tfrac{1}{16}\cot\theta + C.\end{aligned}$$

We must now return to the original variable of integration, x. Since $\theta = \arcsin(x/4)$, we could write $-\frac{1}{16}\cot\theta$ as $-\frac{1}{16}\cot\arcsin(x/4)$, but this is a cumbersome expression. Since the integrand contains $\sqrt{16-x^2}$, it is preferable that the evaluated form also contain this radical. There is a simple geometric method for ensuring that it does. If $0 < \theta < \pi/2$ and $\sin\theta = x/4$, we may interpret θ as an acute angle of a right triangle having opposite side and hypotenuse of lengths x and 4, respectively (see Figure 7.1). By the Pythagorean theorem, the length of the adjacent side is $\sqrt{16-x^2}$. Referring to the triangle, we find

$$\cot\theta = \frac{\sqrt{16-x^2}}{x}.$$

Figure 7.1

$\sin\theta = \dfrac{x}{4}$



It can be shown that the last formula is also true if $-\pi/2 < \theta < 0$. Thus, Figure 7.1 may be used if θ is either positive or negative.

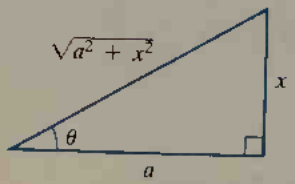
Substituting $\sqrt{16-x^2}/x$ for $\cot\theta$ in our integral evaluation gives us

$$\int \frac{1}{x^2\sqrt{16-x^2}}\,dx = -\frac{1}{16}\cdot\frac{\sqrt{16-x^2}}{x} + C$$
$$= -\frac{\sqrt{16-x^2}}{16x} + C.$$

If an integrand contains $\sqrt{a^2+x^2}$ for $a > 0$, then, by (7.4), we use the substitution $x = a\tan\theta$ to eliminate the radical. When using this substitution, we assume that θ is in the range of the inverse tangent function—that is, $-\pi/2 < \theta < \pi/2$. In this case, $\sec\theta > 0$ and

Figure 7.2

$\tan\theta = \dfrac{x}{a}$



$$\begin{aligned}\sqrt{a^2+x^2} &= \sqrt{a^2+a^2\tan^2\theta}\\ &= \sqrt{a^2(1+\tan^2\theta)}\\ &= \sqrt{a^2\sec^2\theta}\\ &= a\sec\theta.\end{aligned}$$

After substituting and evaluating the resulting trigonometric integral, it is necessary to return to the variable x. We can do so by using the formula $\tan\theta = x/a$ and referring to the right triangle shown in Figure 7.2.

EXAMPLE 2 Evaluate $\displaystyle\int \frac{1}{\sqrt{4+x^2}}\,dx$.

SOLUTION The denominator of the integrand has the form $\sqrt{a^2+x^2}$ with $a = 2$. Hence, by (7.4), we make the substitution

$$x = 2\tan\theta, \qquad dx = 2\sec^2\theta\,d\theta.$$

Consequently,

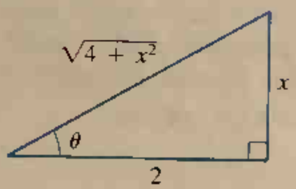
$$\begin{aligned}\sqrt{4+x^2} &= \sqrt{4+4\tan^2\theta}\\ &= 2\sqrt{1+\tan^2\theta} = 2\sqrt{\sec^2\theta} = 2\sec\theta\end{aligned}$$

and

$$\begin{aligned}\int \frac{1}{\sqrt{4+x^2}}\,dx &= \int \frac{1}{2\sec\theta}2\sec^2\theta\,d\theta\\ &= \int \sec\theta\,d\theta\\ &= \ln|\sec\theta+\tan\theta| + C.\end{aligned}$$

Figure 7.3

$\tan\theta = \dfrac{x}{2}$



Using $\tan\theta = x/2$, we sketch the triangle in Figure 7.3, from which we obtain

$$\sec\theta = \frac{\sqrt{4+x^2}}{2}.$$

Hence,

$$\int \frac{1}{\sqrt{4+x^2}}\,dx = \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C.$$

The expression on the right may be written

$$\ln\left|\frac{\sqrt{4+x^2}+x}{2}\right| + C = \ln\left|\sqrt{4+x^2}+x\right| - \ln 2 + C.$$

Since $\sqrt{4+x^2}+x > 0$ for every x, the absolute value sign is unnecessary. If we also let $D = -\ln 2 + C$, then

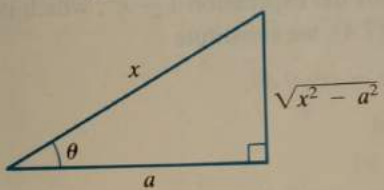
$$\int \frac{1}{\sqrt{4+x^2}}\,dx = \ln\left(\sqrt{4+x^2}+x\right) + D.$$

If an integrand contains $\sqrt{x^2-a^2}$, then using (7.4) we substitute $x = a\sec\theta$, where θ is chosen in the range of the inverse secant function—that is, either $0 \le \theta < \pi/2$ or $\pi \le \theta < 3\pi/2$. In this case, $\tan\theta \ge 0$ and

$$\begin{aligned}\sqrt{x^2-a^2} &= \sqrt{a^2\sec^2\theta - a^2}\\ &= \sqrt{a^2(\sec^2\theta - 1)}\\ &= \sqrt{a^2\tan^2\theta}\\ &= a\tan\theta.\end{aligned}$$

Figure 7.4

$\sec\theta = \dfrac{x}{a}$



Since

$$\sec\theta = \frac{x}{a},$$

we may refer to the triangle in Figure 7.4 when changing from the variable θ to the variable x.

EXAMPLE ▪ 3 Evaluate $\displaystyle\int \frac{\sqrt{x^2-9}}{x}\,dx.$

SOLUTION The integrand contains $\sqrt{x^2-9}$, which is of the form $\sqrt{x^2-a^2}$ with $a = 3$. Referring to (7.4), we substitute as follows:

$$x = 3\sec\theta, \qquad dx = 3\sec\theta\tan\theta\,d\theta$$

Consequently,

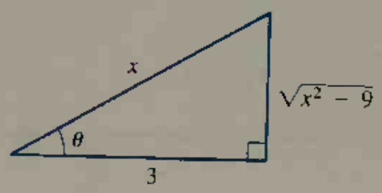
$$\begin{aligned}\sqrt{x^2-9} &= \sqrt{9\sec^2\theta - 9}\\ &= 3\sqrt{\sec^2\theta - 1} = 3\sqrt{\tan^2\theta} = 3\tan\theta\end{aligned}$$

and

$$\int \frac{\sqrt{x^2-9}}{x}\,dx = \int \frac{3\tan\theta}{3\sec\theta} 3\sec\theta\tan\theta\,d\theta$$
$$= 3\int \tan^2\theta\,d\theta$$
$$= 3\int (\sec^2\theta - 1)\,d\theta = 3\int \sec^2\theta\,d\theta - 3\int d\theta$$
$$= 3\tan\theta - 3\theta + C.$$

Figure 7.5

$\sec\theta = \dfrac{x}{3}$



Since $\sec\theta = x/3$, we may refer to the right triangle in Figure 7.5. Using $\tan\theta = \sqrt{x^2-9}/3$ and $\theta = \sec^{-1}(x/3)$, we obtain

$$\int \frac{\sqrt{x^2-9}}{x}\,dx = 3\frac{\sqrt{x^2-9}}{3} - 3\sec^{-1}\left(\frac{x}{3}\right) + C$$
$$= \sqrt{x^2-9} - 3\sec^{-1}\left(\frac{x}{3}\right) + C.$$

As shown in the next example, we can use trigonometric substitutions to evaluate certain integrals that involve $(a^2-x^2)^r$, $(a^2+x^2)^r$, or $(x^2-a^2)^r$, in cases other than $r = \frac{1}{2}$.

EXAMPLE ▪ 4 Evaluate $\displaystyle\int \frac{(1-x^2)^{3/2}}{x^6}\,dx$.

SOLUTION The integrand contains the expression $1-x^2$, which is of the form a^2-x^2 with $a=1$. Using (7.4), we substitute

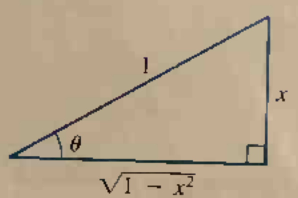
$$x = \sin\theta, \qquad dx = \cos\theta\,d\theta.$$

Thus, $1-x^2 = 1-\sin^2\theta = \cos^2\theta$, and

$$\int \frac{(1-x^2)^{3/2}}{x^6}\,dx = \int \frac{(\cos^2\theta)^{3/2}}{\sin^6\theta}\cos\theta\,d\theta$$
$$= \int \frac{\cos^4\theta}{\sin^6\theta}\,d\theta = \int \frac{\cos^4\theta}{\sin^4\theta}\cdot\frac{1}{\sin^2\theta}\,d\theta$$
$$= \int \cot^4\theta\csc^2\theta\,d\theta$$
$$= -\tfrac{1}{5}\cot^5\theta + C.$$

Figure 7.6

$\sin\theta = x$



To return to the variable x, we note that $\sin\theta = x = x/1$ and refer to the right triangle in Figure 7.6, obtaining $\cot\theta = \sqrt{1-x^2}/x$. Hence,

$$\int \frac{(1-x^2)^{3/2}}{x^6}\,dx = -\frac{1}{5}\left(\frac{\sqrt{1-x^2}}{x}\right)^5 + C = -\frac{(1-x^2)^{5/2}}{5x^5} + C.$$

Trigonometric substitutions may also be used with trigonometric identities in the evaluation of definite integrals. In the next example, we use the substitution $x = a \sin\theta$ and the identity $2\cos^2\theta = 1 + \cos 2\theta$ to find the area bounded by an ellipse.

EXAMPLE ▪ 5 Find the area of the region bounded by an ellipse whose major and minor axes have lengths $2a$ and $2b$, respectively.

SOLUTION By Theorem 34 (page 70), we see that an equation for the ellipse is $(x^2/a^2) + (y^2/b^2) = 1$. Solving for y gives us

$$y = \pm\frac{b}{a}\sqrt{a^2 - x^2}.$$

The graph of the ellipse has the general shape shown in Figure 64 (page 69) and hence, by symmetry, it is sufficient to find the area of the region in the first quadrant and multiply the result by 4. By Theorem (4.19),

$$A = 4\int_0^a y\,dx = 4\frac{b}{a}\int_0^a \sqrt{a^2 - x^2}\,dx.$$

If we make the trigonometric substitution $x = a\sin\theta$, then

$$\sqrt{a^2 - x^2} = a\cos\theta \quad \text{and} \quad dx = a\cos\theta\,d\theta.$$

Since the values of θ that correspond to $x = 0$ and $x = a$ are $\theta = 0$ and $\theta = \pi/2$, respectively, we obtain

$$\begin{aligned} A &= 4\frac{b}{a}\int_0^{\pi/2} a^2\cos^2\theta\,d\theta = 4ab\int_0^{\pi/2}\frac{1+\cos 2\theta}{2}\,d\theta \\ &= 2ab\left[\theta + \tfrac{1}{2}\sin 2\theta\right]_0^{\pi/2} \\ &= 2ab\left[\frac{\pi}{2}\right] = \pi ab. \end{aligned}$$

Thus, the area of an ellipse with axes of lengths $2a$ and $2b$ is πab. As a special case, if $b = a$, the ellipse is a circle and $A = \pi a^2$.

Although we now have additional integration techniques available, it is a good idea to keep earlier methods in mind. For example, the integral $\int (x/\sqrt{9 + x^2})\,dx$ could be evaluated by means of the trigonometric substitution $x = 3\tan\theta$. However, it is simpler to use the algebraic substitution $u = 9 + x^2$ and $du = 2x\,dx$, for in this event the integral takes on the form $\frac{1}{2}\int u^{-1/2}\,du$, which is readily integrated by means of the power rule. The following exercises include integrals that can be evaluated using simpler techniques than trigonometric substitutions.

EXERCISES 7.3

Exer. 1–22: Evaluate the integral.

1 $\displaystyle\int \frac{1}{x\sqrt{4-x^2}}\,dx$

2 $\displaystyle\int \frac{\sqrt{4-x^2}}{x^2}\,dx$

3 $\displaystyle\int \frac{1}{x\sqrt{9+x^2}}\,dx$

4 $\displaystyle\int \frac{1}{x^2\sqrt{x^2+9}}\,dx$

5 $\displaystyle\int \frac{1}{x^2\sqrt{x^2-25}}\,dx$

6 $\displaystyle\int \frac{1}{x^3\sqrt{x^2-25}}\,dx$

7 $\displaystyle\int \frac{x}{\sqrt{4-x^2}}\,dx$

8 $\displaystyle\int \frac{x}{x^2+9}\,dx$

9 $\displaystyle\int \frac{1}{(x^2-1)^{3/2}}\,dx$

10 $\displaystyle\int \frac{1}{\sqrt{4x^2-25}}\,dx$

11 $\displaystyle\int \frac{1}{(36+x^2)^2}\,dx$

12 $\displaystyle\int \frac{1}{(16-x^2)^{5/2}}\,dx$

13 $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

14 $\displaystyle\int \frac{1}{49+x^2}\,dx$

15 $\displaystyle\int \frac{x}{(16-x^2)^2}\,dx$

16 $\displaystyle\int x\sqrt{x^2-9}\,dx$

17 $\displaystyle\int \frac{x^3}{\sqrt{9x^2+49}}\,dx$

18 $\displaystyle\int \frac{1}{x\sqrt{25x^2+16}}\,dx$

19 $\displaystyle\int \frac{1}{x^4\sqrt{x^2-3}}\,dx$

20 $\displaystyle\int \frac{x^2}{(1-9x^2)^{3/2}}\,dx$

21 $\displaystyle\int \frac{(4+x^2)^2}{x^3}\,dx$

22 $\displaystyle\int \frac{3x-5}{\sqrt{1-x^2}}\,dx$

23 The region bounded by the graphs of $y = 0$, $x = 5$, and $y = x(x^2+25)^{-1/2}$ is revolved about the y-axis. Find the volume of the resulting solid.

24 Find the area of the region bounded by the graph of $y = x^3(10-x^2)^{-1/2}$, the x-axis, and the line $x = 1$.

25 The shape of the earth's surface can be approximated by revolving the ellipse $(x^2/a^2) + (y^2/b^2) = 1$, with $a = 6378$ km and $b = 6356$ km, about the x-axis. Approximate the surface area of the earth to the nearest 10^6 km^2. (*Hint:* Use (5.19) with $f(x) = \sqrt{b^2 - (b^2/a^2)x^2}$, and make the substitution $u = (b/a)x$.)

26 Let R be the region bounded by the right branch of the hyperbola $x^2 - y^2 = 8$ and the vertical line through the focus. Find the area of the curved surface of the solid obtained by revolving R about the x-axis.

Exer. 27–28: Solve the differential equation subject to the given initial condition.

27 $x\,dy = \sqrt{x^2-16}\,dx;\quad y = 0$ if $x = 4$

28 $\sqrt{1-x^2}\,dy = x^3\,dx;\quad y = 0$ if $x = 0$

Exer. 29–34: Use a trigonometric substitution to derive the formula. (See Formulas 21, 27, 31, 36, 41, and 44 in Appendix II.)

29 $$\int \sqrt{a^2+u^2}\,du = \frac{u}{2}\sqrt{a^2+u^2} + \frac{a^2}{2}\ln\left|u+\sqrt{a^2+u^2}\right| + C$$

30 $$\int \frac{1}{u\sqrt{a^2+u^2}}\,du = -\frac{1}{a}\ln\left|\frac{\sqrt{a^2+u^2}+a}{u}\right| + C$$

31 $$\int u^2\sqrt{a^2-u^2}\,du = \frac{u}{8}(2u^2-a^2)\sqrt{a^2-u^2} + \frac{a^4}{8}\sin^{-1}\frac{u}{a} + C$$

32 $$\int \frac{1}{u^2\sqrt{a^2-u^2}}\,du = -\frac{1}{a^2u}\sqrt{a^2-u^2} + C$$

33 $$\int \frac{\sqrt{u^2-a^2}}{u}\,du = \sqrt{u^2-a^2} - a\sec^{-1}\frac{u}{a} + C$$

34 $$\int \frac{u^2}{\sqrt{u^2-a^2}}\,du = \frac{u}{2}\sqrt{u^2-a^2} + \frac{a^2}{2}\ln\left|u+\sqrt{u^2-a^2}\right| + C$$

7.4 INTEGRALS OF RATIONAL FUNCTIONS

Recall that if q is a rational function, then $q(x) = f(x)/g(x)$, where $f(x)$ and $g(x)$ are polynomials. In this section, we examine the rules for evaluating $\int q(x)\,dx$.